

Windows Platform 1

Creating and managing Network using Windows Server 2003

الأهداف:

تهدف هذه الوظيفة إلى دراسة مجموعة من الحالات الافتراضية التي يمكن أن يتعرض لها مدير شبكة تعمل باستخدام نظام Windows Server 2003 واقتراح الحلول المناسبة لهذه الحالات وتنفيذ هذه الحلول وتوثيقها بشكل مناسب لتقديمها كحل لهذه الوظيفة. بالإضافة إلى الإجابة على بعض الاسئلة التي قد تستدعي اختبار بعض الوظائف في نظام ويندوز.



Windows Server® 2008 R2

INTRODUCTION:	3
HOMEWORK TEXT:	3
QUESTION 1:	3
QUESTION 2:	3
QUESTION 3:	5
QUESTION 4:	6
QUESTION 5:	6
QUESTION 6:	6
QUESTION 7:	7
QUESTION 8:	7
QUESTION 9:	8
QUESTION 10:	8
QUESTION 11:	8
QUESTION 12:	9
QUESTION 13:	9
QUESTION 14:	9
QUESTION 15:	10
QUESTION 16:	10
QUESTION 17:	11
QUESTION 18:	12
QUESTION 19:	12
QUESTION 20:	13
QUESTION 21:	13
QUESTION 22:	14
QUESTION 23:	15
QUESTION 24:	15
QUESTION 25:	16
QUESTION 26:	16
QUESTION 27:	16
QUESTION 28:	17
QUESTION 29:	17
QUESTION 30:	17
QUESTION 31:	17
GROUP MEMBERS:	17



Introduction:

The Windows Server 2003 (also referred to as Win2K3) is a server operating system produced by Microsoft. Windows Server 2003 includes technology to accelerate server and Active Directory (AD) deployments, dramatically improved migration tools (especially for NT upgrades), a best-of-breed (if poorly defined) application server, a new Group Policy Management Console (GPMC), improved file and print services, task- and role-based management tools, numerous security improvements including various feature lock-downs, dramatic performance improvements when compared to Windows 2000 Server on the same hardware, a Volume Shadow Copies feature that stores file snapshots, and enhancements to virtually every subsystem imaginable, including Windows Media Services, IIS, Terminal Services, and the like.

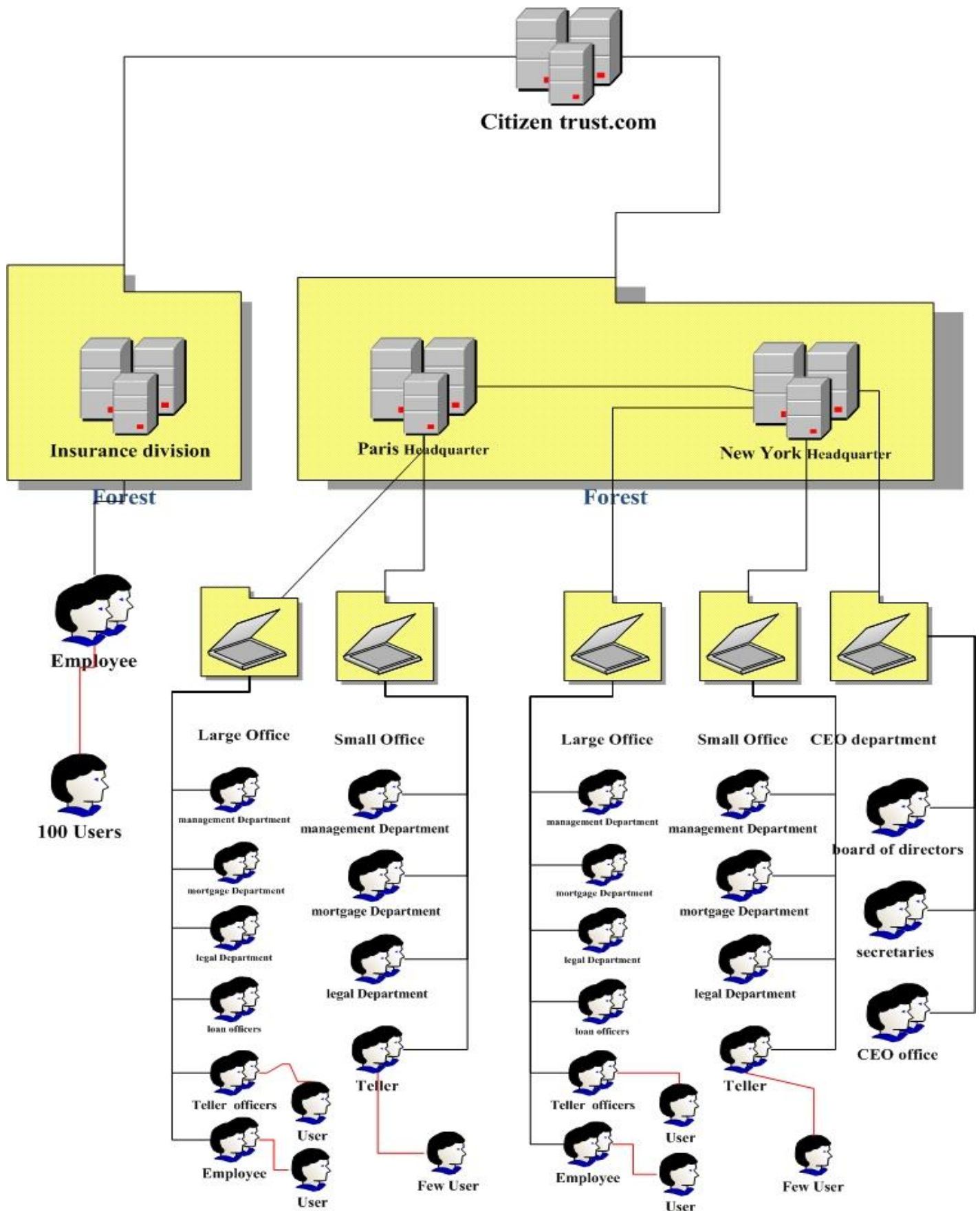
Homework text:

Citizens State Trust is a large international bank with a North American headquarters in New York and a European headquarters in Paris. They have branch offices throughout North America and Europe. Some branch offices are large, with hundreds of tellers, loan officers, and other employees. Others are smaller, with only a few tellers. The bank also has a new insurance division, which operates out of San Francisco and has only 100 employees. While the Insurance division is owned by Citizens State Trust, it is managed independently of the bank. Each branch, in addition to the above mentioned, has management department, mortgage department, legal department, and the North American headquarters, has CEO department, which includes board of directors, secretaries, and CEO office.

Question 1:

Imagine that you are involved in the early stages of designing the IT infrastructure for the network of this international bank, and you are asked to provide a proposal to a third party. Present your design in a graphical format; suggest Domains, Groups, Users, OUs (organizational Units), to allow good functionality and lowest TCO (Total Cost of Ownership) for the implementation. (10 pts)

Microsoft®
Windows Server 2003
Enterprise Edition



Question 2:

Why would you want to use Active Directory? What advantages does Active Directory offer over using local user accounts on standalone servers? (3 pts)

- Active Directory® is the Microsoft's implementation of Directory Services its purpose is to store information about users, resources, and other network components, and to provide that information according to access permissions of the entity who is requesting it.
- Centralization: Active Directory is centralized directory implementation providing a single database of network resources.
- Scalability: Active Directory allows its database to be partitioned and distributed across the domains that make the network. But still Active Directory can be managed as a single directory.
- Standardization: Active Directory is standardized because it is made accessible through Lightweight Directory Access Protocol (LDAP), which is an IETF standard.
- Extensible: Active Directory is extensible. It allows third-party developers to store the information of their own application inside Active Directory and make use of the features provided by the Active Directory.
- Separation of physical network: Active Directory makes the physical structure of the network transparent and only the local structure is visible to the users.
- Security: Active Directory is tightly integrated with the Windows 2003 server security and the major security protocols make it more secure.
- Domain Name System (DNS) support: The Active Directory supports Domain Name System (DNS) and Active Directory requires DNS to function properly.
- TCP/IP compatibility: Active Directory and Windows Server 2003 utilize the TCP/IP protocol stack as their primary method of communications.

Question 3:

Suppose that a windows server 2003 web edition exists on one of pre-owned boxes that the bank is thinking to acquire. Would it possible to setup AD and mange user access to files and set permission on that machine, since the goal is to make that box a Domain Controller? (3 pts)

No, we can't. Web Edition cannot act as a domain controller.

Question 4:

How Many DCs (Domain Controllers) do you need, and why? (3 pts)

3 Domain (Headquarter in New York, Headquarter in Paris, insurance division which is managed independently of the bank)

Question 5:

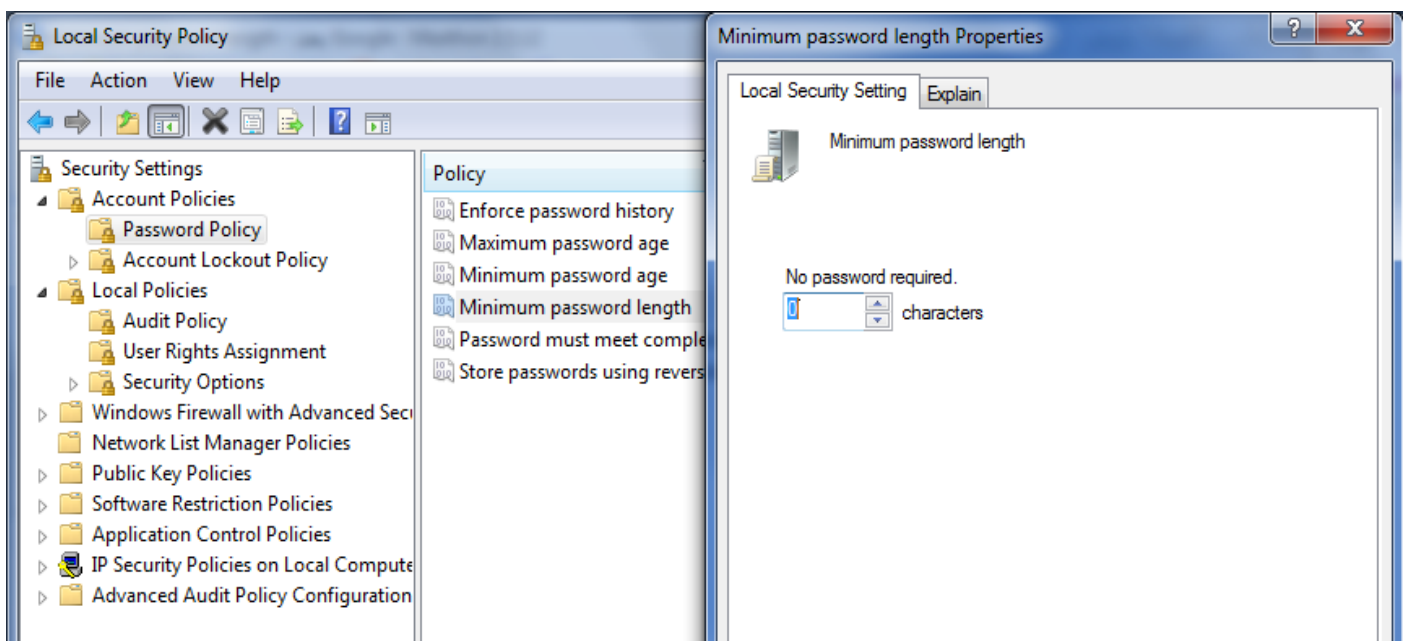
What process allows domain controllers to synchronize their copies of the directory? Explain? (3 pts)

Using Active Directory Replication, Windows 2003 Active Directory has a distributed directory structure and stores objects (users, computers, printers etc). Objects which are stored in Active Directory are distributed different domain controllers in a forest. Active Directory replication is the process by which the changes that originate on one domain controller are automatically transferred to other domain controllers in the forest.

Question 6:

How can you force users to select passwords with at least ten characters? (3 pts)

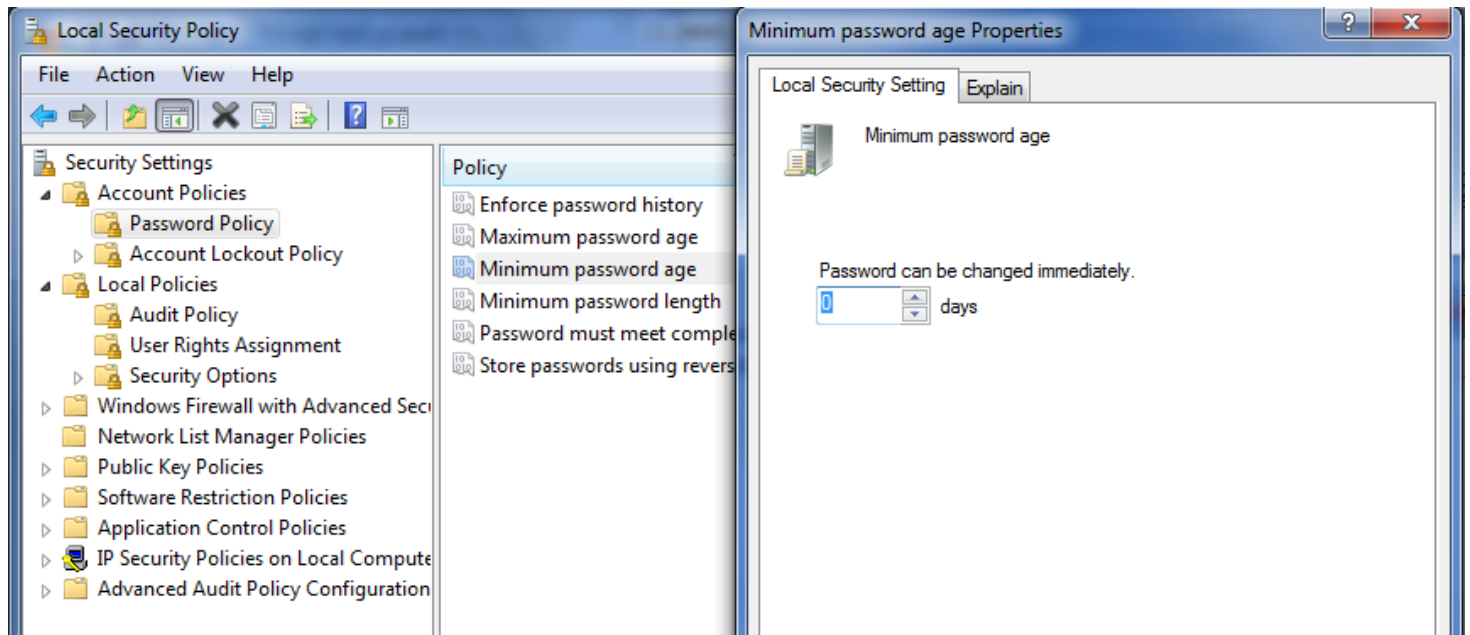
By assigning value: 10 to Minimum password length from Local Security Policy: Password policy.



Question 7:

How can you prevent users from changing their passwords and then immediately changing back to their old passwords? (3 pts)

By assigning value to Minimum password age from Local Security Policy: Password policy.



Question 8:

What is the difference between a standalone server and a member server? (3 pts)

Standalone server: A Standalone server is a type of server that is not part of a larger network infrastructure such as an Active Directory domain. In a general sense, a standalone server provides services to clients without relying on any external resources, including authentication. However, in the Windows Server System, a server can either be a Domain Controller, a Member server (joined to a domain), or a Standalone server (joined to a workgroup). Therefore, a single server that is hosting an Active Directory domain would be considered a Domain Controller and not a standalone server.

Member server: A member server is a computer that runs an operating system in the Windows 2000 Server family or the Windows Server 2003 family, belongs to a domain, and is not a domain controller. Member servers typically function as the following types of servers: file servers, application servers, database servers, Web servers, certificate servers, firewalls and remote-access servers.

Question 9:

How can you remove Active Directory from a domain controller and make it a standalone server again? (3 pts)

Dcpromo is a utility which comes with Windows 2003 Server products which is used to Install and uninstall Active Directory Services. To run dcpromo, go to Start > Run, type dcpromo



Question 10:

When is a single domain appropriate for an organization? (3 pts)

A single domain is appropriate for an organization if it want to attain centralized administration, which accomplished through the use of OUs and Group Policy objects, and minimize headaches in the process. So a single domain model allows for a high level of administrative control and the ability to delegate tasks to lower sets of administrators. Also, single domain appropriate to centralized management of user accounts when the number of user is relatively low

Question 11:

What must a DNS server provide in order to be compatible with Active Directory? (3 pts)

Active Directory is fully integrated with DNS and TCP/IP—DNS is required. To be fully functional, the DNS server must support SRV resource records or service records. In order for Active Directory to function properly, DNS servers must provide support for Service Location (SRV) resource records described in RFC 2052, A DNS RR for specifying the location of services (DNS SRV). SRV resource records map the name of a service to the name of a server offering that service. Active Directory clients and domain controllers use SRV records to determine the IP addresses of domain controllers. Although not a technical requirement of Active Directory, it is highly recommended that DNS servers provide support for DNS dynamic updates described in RFC 2136, Observations on the use of Components of the Class A Address Space within the Internet



Question 12:

What is the best method or methods to install Windows Server 2003 on a dozen identical computers that are attached to your network? (3 pts)

- Automated Installation is the best method to install Windows Server 2003 repeatedly on many computers with the same hardware configurations.
- Across the network installation: you can perform this if the Windows 2003 installation files are available on some other computer attached to the network.
- Remote installation: You can do this with Microsoft's Remote Installation Services (RIS), which enables a network to push a Windows 2003 installation out to network systems.
- Using some utilities such as Norton Ghost which can take an image of one of them (12 Computers) and spread it on the rest of PCs.

Question 13:

In such large company you're likely to find a number of different drive-usage scenarios, give an example of fault tolerance disk drives for a Windows Server 2003 computer that contains the office's files and provides shared printing for the office's users? (3 pts)

RAID is commonly implemented for both performance and fault tolerance. With RAID, you can choose to assemble disks to provide fault tolerance, performance, or both, depending on the RAID level that you configure. And it applies To: Windows Server 2003, Windows Server 2003 with SP1 and Windows Server 2003 with SP2

Question 14:

What file system you would use? What are the advantages of the NTFS file system? (3 pts)

It's always preferred to use the NTFS file system as often as possible.

Advantages of NTFS

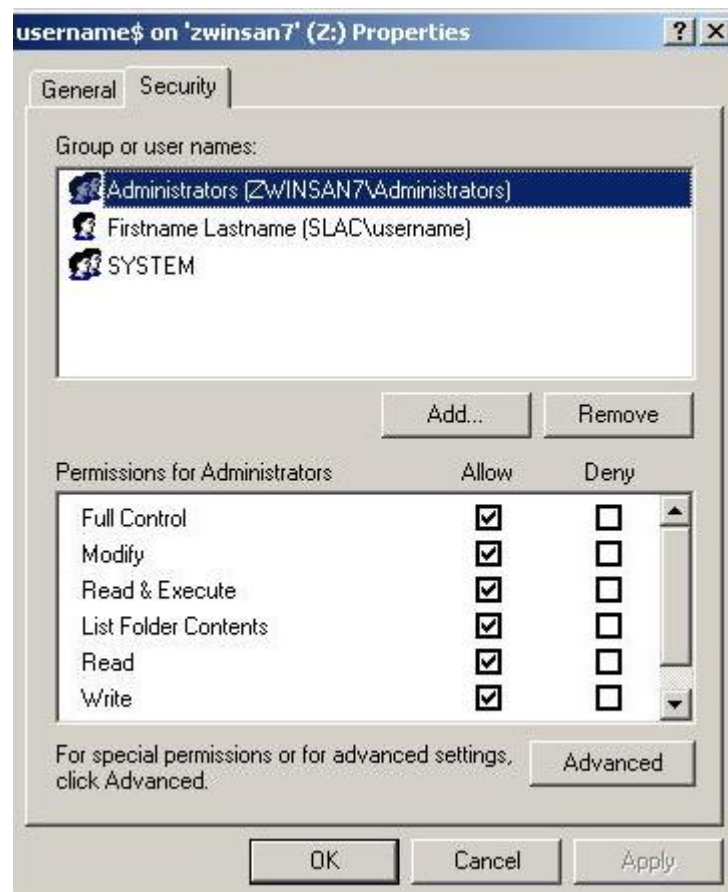
- NTFS is a recoverable file system.
- NTFS supports compression on volumes, folders, and files. Files that are compressed on an NTFS volume can be read and written by any Windows-based application without first being decompressed by another program
- NTFS manages disk space more efficiently than FAT, using smaller clusters (4 KB for volumes up to 2 terabytes).
- NTFS does not restrict the number of entries in the root folder.
- The boot sector is backed up to a sector at the end of the volume.

- NTFS minimizes the number of disk accesses required to find a file.
- NTFS supports a native encryption system, EFS, that uses symmetric key encryption in conjunction with public key technology to prevent unauthorized access to file contents.
- NTFS supports sparse files so that very large files can be written to disk while requiring only a small amount of storage space.

Question 15:

What permission allows a user to read, write, and change permissions on a file? (3 pts)

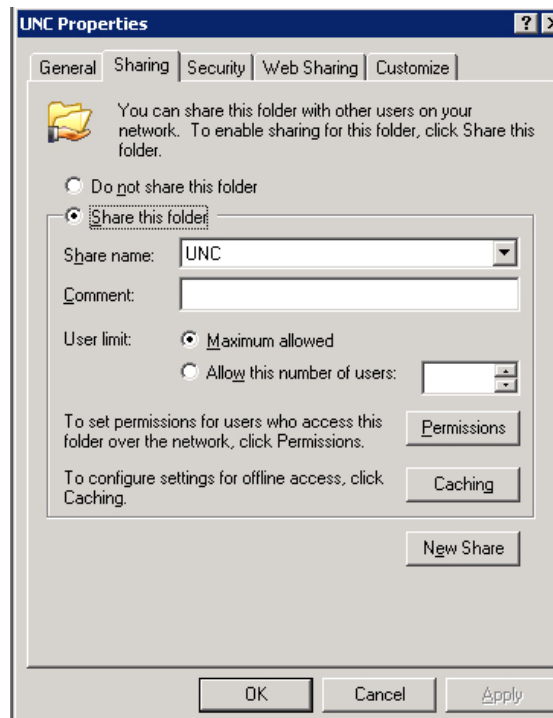
Full control. This allows the user or group to change permissions on the folder, take ownership of it, and perform all activities included in all other permissions.



Question 16:

How can you make files accessible to users on the network? (3 pts)

By enable sharing on files. Local Computer files can be shared using Windows Explorer. To share a file, right click the file and select "Properties" from the context menu. Click the "Sharing" tab.

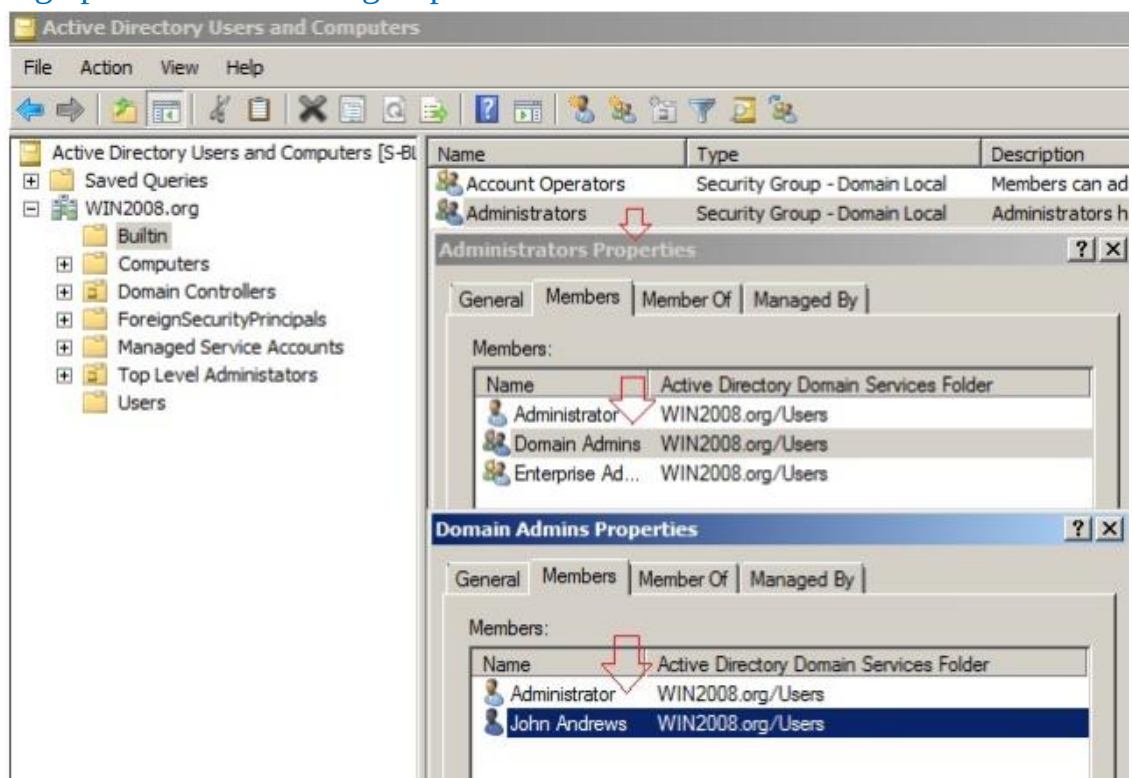


Question 17:

In what order should you place users in groups and assign file permissions? (3 pts)

First: Add them in Domain Local group.

Second: Assign permissions to the group.



Question 18:

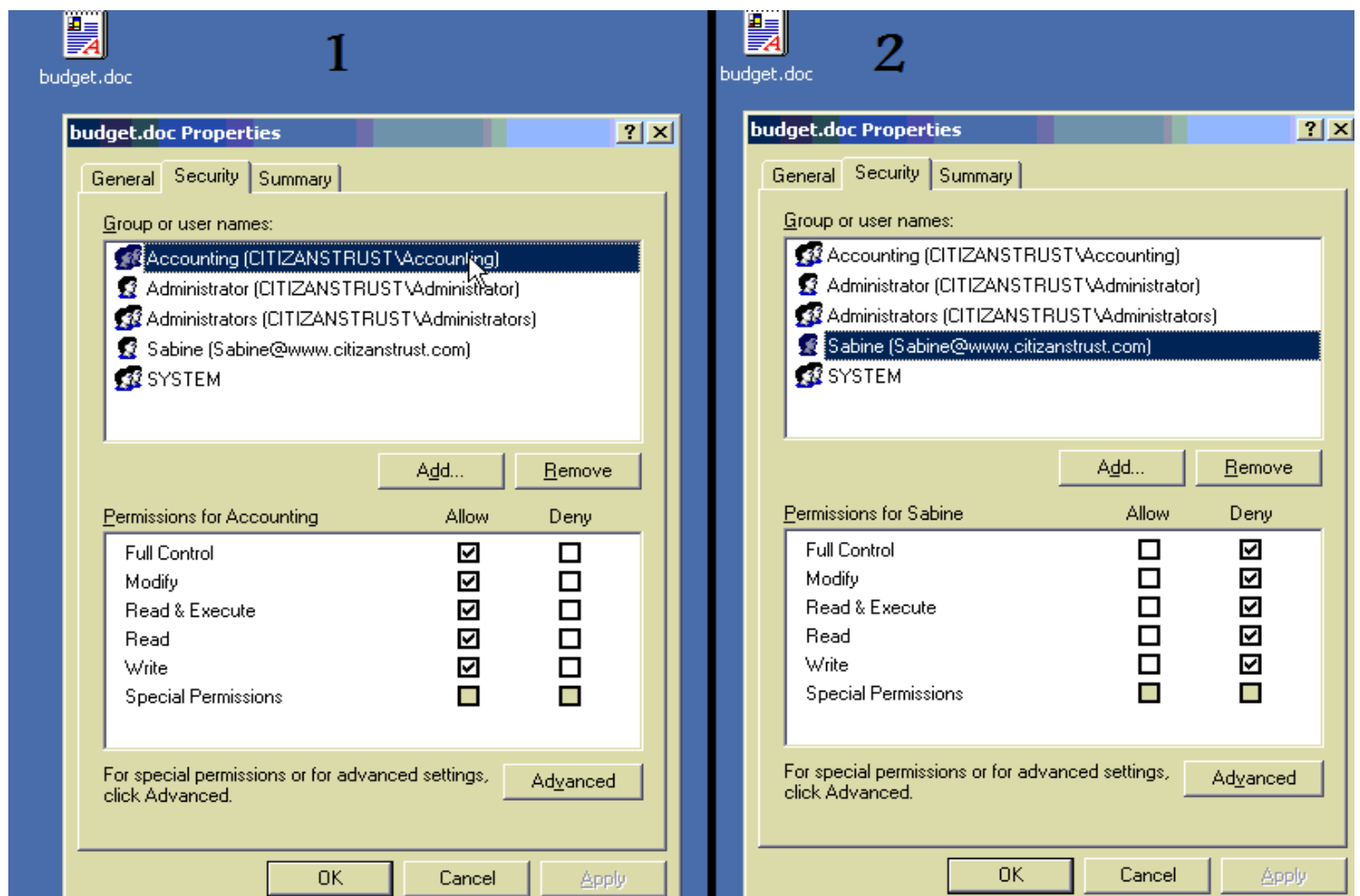
If a group called TexasClients is used to sending marketing emails to the clients in Texas what type of groups would you use? (3 pts)

Distribution groups, Can be used only with e-mail applications (such as Exchange) to send e-mail to collections of users.

Question 19:

Imagine that you work for Citizens State Trust and have a user named Sabine. Sabine belongs to a group named Accounting. How would you deny Sabine access to a file: budget.doc and let the rest of the Accounting employees have access to it? (3 pts)

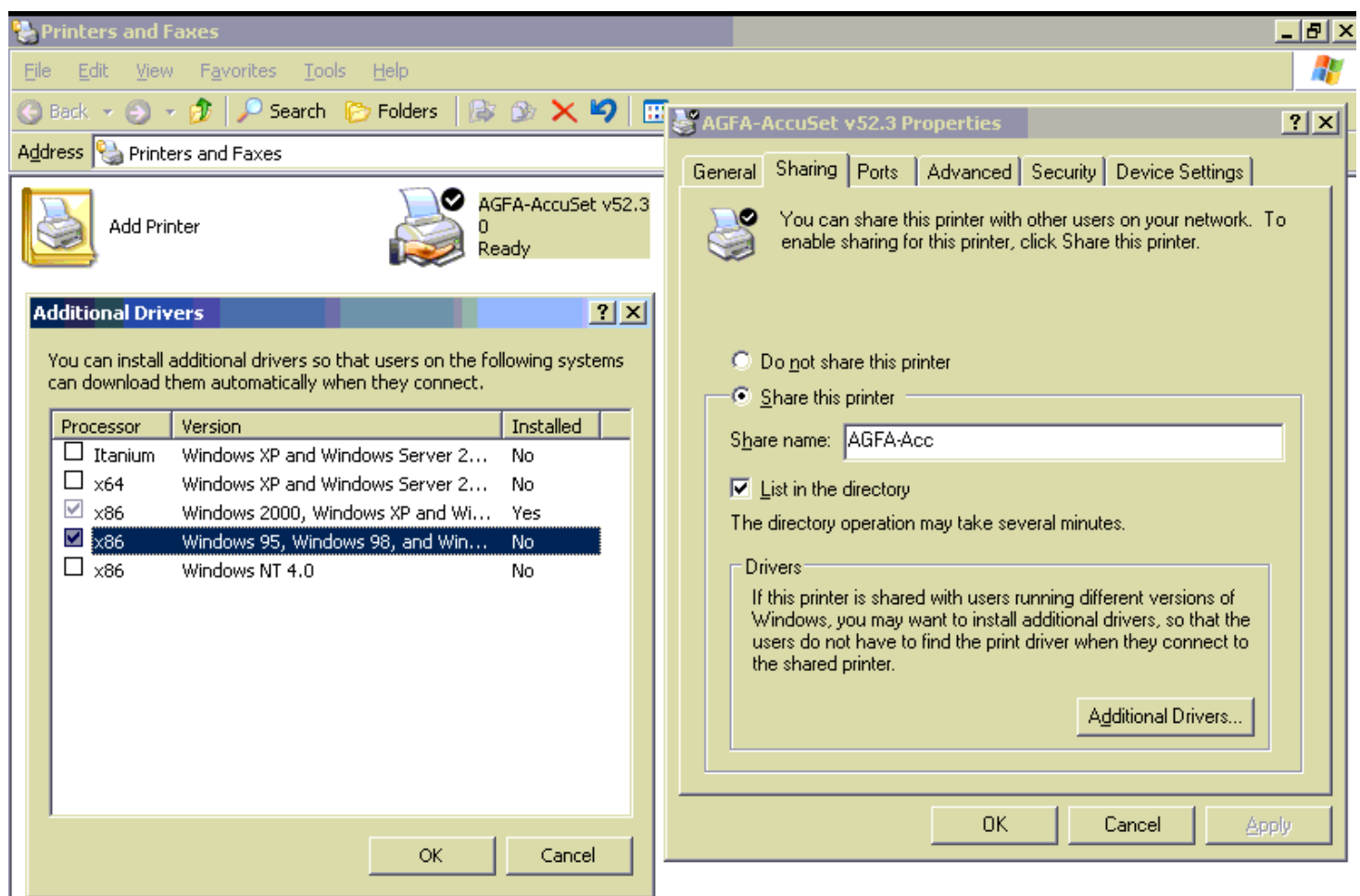
Right click on the folder which you want to block from other users and select Properties. Now click the Security tab in the folder properties window, then, Click on the Add button to open the “Select Users or Group” window.



Question 20:

Citizens State Trust plans to use networked print devices in all of its offices and set up printers on each office's Windows Server 2003 computer. Each office will include one dot-matrix print device for each teller window and will also include three laser print devices. If your environment includes Windows 98 client computers, how can you ensure that those clients are able to print to the printer you share on your Windows Server 2003 computer without requiring additional CDs or floppy disks? (3 pts)

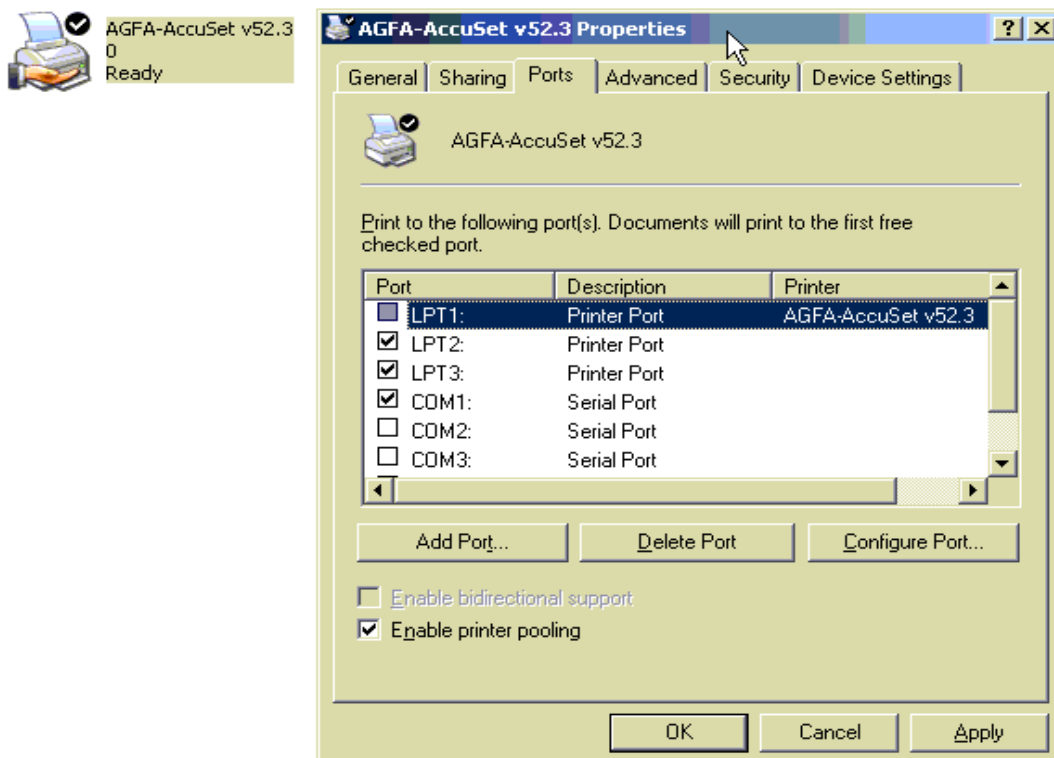
After install printer drivers on windows 2003 server, Right click on the printer (located in printer and faxes), and select Properties. Now click the sharing tab in the selected printer properties window, then, Click on the additional drivers button to open the "additional driver" window. At the end check the (x86 windows95, windows98, and win...) then, press ok.



Question 21:

In such a busy environment, how can you combine several identical print devices so that they balance your organization's printing workload between them? (3 pts)

Right click on the printer (located in printer and faxes), and select Properties. Now click the Port tab in the selected printer properties window, and then check the Enable Printer Pooling. Then check ports that you need together you can choose more than one port. Finally, press ok.



Question 22:

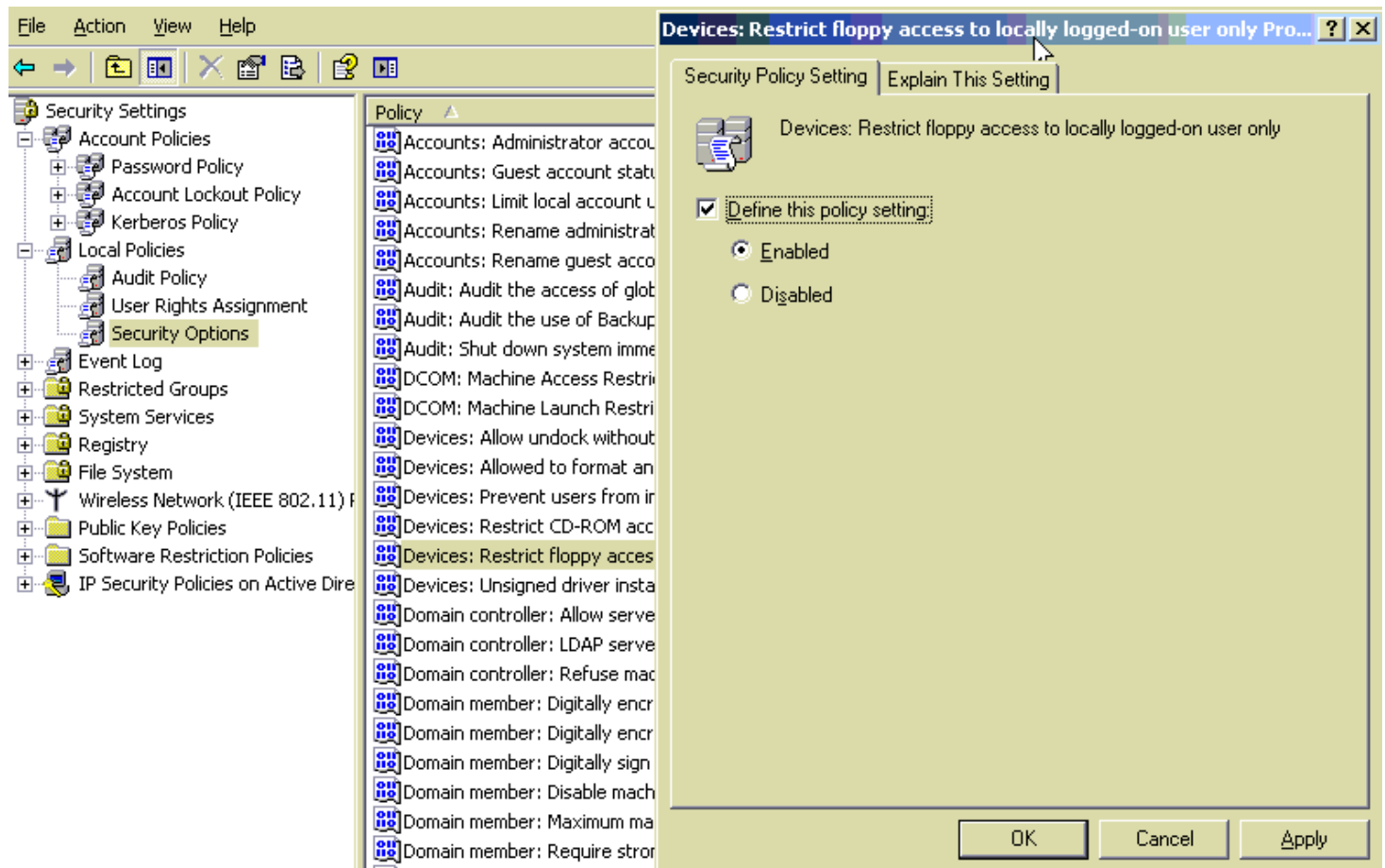
Citizens State Trust plans to use aggressive security policies to protect their computers. They plan to apply policies regarding password length, account lockout, and password age to all computers, ensuring that even local computer accounts will be more secure. List 4 security best practices that the bank can implement? (3 pts)

- Apply (a minimum password of six characters, Maximum password age between 45 and 60 days , Minimum password age between 3 and 5 days).
- Enforce password history 5 or greater so system will remember the last 5 passwords used by a user so he or she can't reuse any of those passwords.
- Make account lock after 3 to 5 times failed logon and make lockout duration about 30 minutes.
- Change Local Administrator name.

Question 23:

How can you keep network users from accessing the contents of a server's floppy drive, even if that drive has been shared? (3 pts)

By enable Restrict floppy access to locally logged-on user only



Question 24:

The Domain Name System (DNS) is one of the key TCP/IP network services required by all TCP/IP-based networks. DNS provides name resolution services, allowing computers to translate human friendly names like www.CST.com into TCP/IP addresses. What tool do you use to administer Windows Server 2003's DNS software?

NSLOOKUP

How to setup static entries and how can you reduce the amount of manual DNS configuration? (3 pts)

By registering the users automatically.

Question 25:

Suppose that older versions of Windows still exist on some pre-owned machines that the bank is thinking to acquire. What can be used as their primary means of name resolution? Explain how to configure that service? (3 pts)

Name resolution is an important building blocks for Microsoft Networking, as with any network service. In fact any network name needs to be resolved to a network address—that is, when you point your browser to [http:// www.svuonline.org](http://www.svuonline.org) , the name gets resolved via DNS (Domain Name Service) to the network address 213.178.230.172. The latest versions of Microsoft networking do just this: they use DNS as their primary means of name resolution and server location. The bad news is that there are a couple of backward-compatible issues that you will need to understand to effectively troubleshoot. Furthermore, even if you are an old hand at DNS, you'll need to learn a bit more about how Microsoft does it. All the methods of Microsoft name resolution and server location (which in aggregate are called the Windows Locator Service) can be summarized as

- NetBIOS standalone (broadcast or static name files)
- NetBIOS point-to-point and flat (non-hierarchical) name server(WINS)
- Hierarchical name server (DNS)

Question 26:

On such a network with thousands of computers, manually configuring the TCP/IP settings of each computer can be very time-consuming. You could easily employ several people full-time just to keep up with TCP/IP configurations—an expensive proposition! What do you suggest in your proposal to help automate TCP/IP configuration? Also give an example on implementing a client? (3 pts)

By using DHCP server. We need to activate (Obtain an IP address automatically) on the Client Computer.

Question 27:

Suggest a plan for disaster recovery, and give an example of implementing this plan? (3 pts)

As most disaster recovery experts will tell you, when it comes to disaster, it's not a matter of if but when the disaster will strike. If you plan ahead and minimize your risks, not only will you sleep better at night, you'll also be able to recover from a disaster that might otherwise have a significant impact on your organization and career. We can't afford to have a down time in our



data centers, and our business-critical services must be available 99.999 percent of the time, so we may opt to utilize third-party disaster recovery solutions and/or services.

- Perform regular backups and keep those backups in a safe and secure place. Ensure that you have a copy of your backup at an off-site location.
- Use geographically dispersed data centers for redundancy to provide business continuity in case of a regional disaster.
- Place your servers in a secure locked room and follow the best practices for securing your hardware. When securing hardware, remember to secure servers, routers, switches, hubs, and other network devices.
- Use uninterruptible power supplies for mission-critical network hardware.
- Monitor your network and services on a regular basis.

Question 28:

What type of backup includes all of the files on your computer that have changed that day?

Daily backup. All selected files and folders that have changed during the day are backed up based on the files' modify date. The archive attribute is neither used nor cleared. If you want to back up all files and folders that change during the day without affecting a backup schedule, use a daily backup.

Question 29:

What tools included with Windows Server 2003 and are used to monitor performance?

Performance tool.

Question 30:

What effect does a full Security Event Log have on your server? (3 pts)

It will slow performance, because, a full security is uses much memory, and processor.

Question 31:

What are the four main areas of a computer that create performance bottlenecks? (3 pts)

▪ CPU	▪ Memory	▪ HDD	▪ I/O
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Group Members:

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